School of Aeronautics and Astronautics

Aeromechanics I

AAE 20300

Instructor Info

Dengfeng Sun

Office Hrs: Tue 10-11am

ARMS 2031

web.ics.purdue.edu/~dsun

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Course Info

Prereq: PHYS 17200

Co-req: MATH 26100

Lab Info

No lab

TA Info

Graduate TAs

Office Hrs: check Brightspace

Undergraduate TAs

Course Description

Fundamental concepts and principles of bodies in motion, with applications to aeronautical and astronautical problems. Subjects covered include rectilinear motion, curvilinear motion, rotation, and plane motion. Static equilibrium and quasi-static equilibrium are treated as a part of motion in which the acceleration is zero.

Learning Resources, Technology

- No textbook is required. A complete set of course notes will be posted on Brightspace. We recommend that you follow the weekly readings listed in the course schedule.
- Weekly homework assignments will be posted on Brightspace and must be submitted through Gradescope.
- For course-related questions, please post on Piazza. Emails to the instructor/GTAs/UTAs regarding questions that can be addressed on Piazza may not receive a response.
- Please use the course Piazza forum actively to stay informed, ask questions, and engage in discussion. On weekdays, one TA will be assigned to monitor Piazza closely. The Piazza link is available on Brightspace.

Learning Outcomes

On completing this course, you shall be able to:

- Compute velocities and accelerations for any type of motion.
- Compute angular velocities and accelerations of rigid bodies.
- Model forces and apply Newton's Laws in particle dynamics problems.
- Model forces and solve general statics of bodies problems.

Attendance Policy

Attendance is strongly encouraged but not required.

Academic Integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace table of contents, under University Policies.

AI Usage

AI tools can be valuable for brainstorming, clarifying concepts, and supporting your learning, and you are welcome to use them in these ways. However, you may not use AI to generate or complete any part of an assignment. Any assignment found to include AI-generated work will receive a score of zero.

Accessibility

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let us know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247. More details are available on our course Brightspace under Accessibility Information.

FAQs

Are there required textbooks?

No textbook is required. A complete set of notes will be uploaded to Brightspace.

Any software requirements?

No.

Any bonus point opportunities?

Yes. Bonus points will be available for completing the end-of-semester course evaluation, with additional opportunities announced later.

Will grades in this course be curved?

Curving is at the discretion of the professor.

Grading Scheme

10%	Homework; weekly assignment, in general.
30%	Exam 1.
30%	Exam 2.
30%	Final Exam, time & location TBD.

Grades will follow the standard scale: A = $[90, 100]$; B = $[80, 90)$; C = $[70, 80)]$; D = $[60, 70)$; F $<$60.

- Weekly homework will be posted on Brightspace. You must upload your completed homework to Gradescope by 11:59 p.m. on the due date.
- You are responsible for ensuring that your homework is submitted correctly by the stated deadline. Gradescope will email you a receipt confirming successful submission. If you do not receive this email, please verify your submission directly on Gradescope.
- You may use one homework extension without a late penalty:
  - You may apply this extension to any one homework assignment.
  - You must notify the instructor of your intent to use the extension at least 24 hours before the original deadline.
  - Extension requests submitted after the original deadline will not be considered.
  - The maximum extension granted is 48 hours beyond the original deadline.
  - After you have used your extension, additional extension requests will not be considered.
  - Exceptions may be made for documented medical emergencies.
- Homework submitted late without an approved extension will receive a score of zero.
- There will be two midterm exams in addition to the final exam. Please refer to the table above for the exam weights used to compute your final grade. Reasonable accommodations may be made for exceptional circumstances that prevent you from taking an exam at the scheduled time; however, requests must be made as early as possible. The instructor reserves the right to exercise discretion, and conflicts will be handled in accordance with college/university policies.
- Notes, textbooks, and crib sheets are not permitted during exams.
- Calculators are not permitted during exams.
- Regrade requests for exams and homework will be accepted for one week after grades are posted.
- Due to limited time and resources, regrade requests will not be accepted for the final exam.
- One problem on the final exam will be based on a homework problem.

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. More detail on the Nondiscrimination Policy Statement is included in Brightspace.

We at the School of Aeronautics and Astronautics are committed to promoting an inclusive environment that recruits, retains, and celebrates the diverse lived-experiences of our students, postdocs, faculty, and staff and to provide them opportunities and training to excel personally and professionally. As aeronautical engineers at Purdue, we are taking intentional steps to act against all forms of racism, sexism, ableism, xenophobia, transphobia, and other structures of oppression embedded in the fabric of our society. We are fostering a diverse and inclusive environment for students, postdocs, staff, and faculty from all underrepresented and underprivileged groups. We understand that equity is a long- term endeavor and affirm our commitment to sustainable efforts that improve the climate and cultural change within our academic institution.

We will actively strive to educate ourselves on the many different ways that systematic exclusion and inequalities present themselves, to recognize and eradicate exclusionary practices, to call out discrimination when we see it, to humbly learn from our missteps, and to commit to building a safe and equitable climate where our faculty, staff, and students can thrive. We commit to actively celebrating, learning from, promoting, and advocating for the diversity of races, religions, political alignment, colors, ages, national origins and ancestries, genetic backgrounds, marital and parental statuses, sexes and sexual orientations, gender identities and expressions, disabilities, and socioeconomic and veteran statuses that individuals bring to our school. Our mission remains the same:

To be the premier source of scientific discoveries and of well-educated aeronautical engineers, fostering strong academic and industrial ties.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack. Sign in and find information and tools at your fingertips, available to you at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am-5 pm.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc. sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is completely free and can be done on BoilerConnect. If you have any questions, please contact Purdue Wellness at evans240@purdue.edu.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours. CAPS also offers resources specific to COVID-19 on its website. Topics range from "Adjusting to the New Normal" to "How to Talk with Professors about Personal Matters."

CARES

The Community, Assistance and Resources for Engineering Students (CARES) Hub is open to support the well-being of all engineering students. The CARES Hub, located in ARMS 1261, is a welcoming, inclusive space for students to study, connect, grab a snack, and relax 7 days per week. There will also be wellness activities, educational workshops, peer mentoring, social events, and on-site counseling. If you need help handling stress or working through a problem, schedule time with our CARES Hub therapist, Jennie Beutler, or drop by ARMS 1251 to speak to Jennie without an appointment during her Let's Talk walk-in hours from 12:30-1:30pm Monday-Friday. Please visit the CARES Hub webpage to learn more. The CARES Hub staff look forward to connecting with you!

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines, and grading percentages may be modified as necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Any relevant changes to this course will be posted on the course website or communicated by contacting the instructor or TAs via email or phone. You are expected to check your `@purdue.edu` email regularly.

Emergency Notification Procedures

Emergency notification procedures follow a simple rule: if you hear a fire alarm *inside*, proceed *outside*. If you hear a siren *outside*, proceed *inside*.

- Indoor fire alarms: Stop class or research and immediately evacuate the building.
  - Proceed to your designated Emergency Assembly Area, away from building doors.
  - Remain outside until police, fire, or other emergency response personnel provide additional guidance or indicate it is safe to leave.
- All Hazards Outdoor Emergency Warning Sirens: Immediately seek shelter (Shelter in Place) in a safe location within the nearest building.
  - "Shelter in place" means seeking immediate shelter inside a building or University residence.
  - This action may be necessary during events such as a tornado, an active threat (e.g., a shooting), or the release of hazardous materials into the outside air.
  - Once safely inside, seek additional details about the emergency and remain in place until police, fire, or other emergency response personnel provide guidance or indicate it is safe to leave.

Additional information: In both cases, you should seek clarifying information through all available means (Campus Emergency Status page, text message, email alerts, TV, radio, etc.). Review the Purdue Emergency Warning Notification System multi-communication layers at:

`http://www.purdue.edu/ehps/emergency_preparedness/warning-system.html`

Emergency Response Procedures

- Review the Emergency Procedures Guidelines:
  `https://www.purdue.edu/emergency_preparedness/flipchart/index.html`
- Review the Building Emergency Plan (available on the Emergency Preparedness website or from the building deputy) for:
  - evacuation routes, exit points, and the emergency assembly area;
  - when and how to evacuate the building;
  - shelter-in-place procedures and locations; and
  - additional building-specific procedures and requirements.

Emergency Preparedness Awareness Video

- *"Run. Hide. Fight.®"* is a 6-minute active shooter awareness video that illustrates what to look for and how to prepare and respond to this type of incident. See:
  `https://www.youtube.com/watch?v=5mzI_5aj4Vs`

More Information

For additional information, refer to the Emergency Preparedness website:

`http://www.purdue.edu/ehps/emergency_preparedness`

All Hazards Warning Sirens

If we are notified of a Shelter in Place requirement due to a tornado warning, we will shelter on the lowest level of this building, away from windows and doors. Preferred locations include basement corridors, basement offices, and basement restrooms.

Class Schedule

Week 1	Introduction	Ch. 1
	Units and dimensions	Ch. 2
	Vectors	Ch. 3.1-3.3
		Homework 1 assigned
Week 2	MLK Day on 1/19	no lecture
	Vectors (cont'd)	Ch. 3.4-3.5
	Kinematics of points	Ch. 4.1-4.2
		Homework 1 due; Homework 2 assigned
Week 3	Kinematics of points (cont'd)	Ch. 4.3-4.5
		Homework 2 due; Homework 3 assigned
Week 4	Kinematics of reference frames / BKE	Ch. 5
		Homework 3 due; Homework 4 assigned
Week 5	General reference frame motions	Ch. 6
	Angular acceleration	Ch. 7
		Homework 4 due; Homework 5 assigned
Week 6	More examples on BKE	Ch. 6-7
	Review for Exam 1	Homework 5 due; Homework 6 assigned
Week 7	Exam 1	
	Kinematic expansions (VE)	Ch. 8.1
Week 8	Kinematic expansions (AE)	Ch. 8.2
	Particle dynamics	Ch. 9.1-9.7
		Homework 6 due; Homework 7 assigned
Week 9	Particle dynamics (cont'd)	Ch. 9.8-9.11
		Homework 7 due; Homework 8 assigned
Week 10	Spring break	no lectures
Week 11	Statics of bodies	Ch. 10.1-10.4
		Homework 8 due; Homework 9 assigned
Week 12	Review for Exam 2	
	Exam 2	

Week 13	Statics of bodies (cont'd)	Ch. 10.5-10.7 Homework 9 due; Homework 10 assigned
Week 14	Statics of bodies (cont'd)	Ch. 10.8-10.9 Homework 10 due; Homework 11 assigned
Week 15	Review	Homework 11 due; Homework 12 assigned (optional; no due)
Week 16	Review	

The instructor will make every effort to follow this schedule; however, due to unforeseen circumstances, it is tentative and subject to change without notice.